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Size and Shape of the Cerebral Cortex in Mammals

I. The Cortical Surface[1]

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Key Words. Cerebral cortex · Brain size · Mammals · Allometry · Evolution · Cortical folding

Abstract. The evolution of the brain in mammals has been accompanied by a progressive enlargement of the cerebral cortex. Allometric analysis of the volume, surface and convolutedness of this cortex shows that among mammals two major groups can be distinguished: (1) species with lissencephalic brains, where surface-volume relationships are determined by linear geometric laws, and (2) species with gyrencephalic brains, where strict geometric similarity no longer applies. Different mathematical models are required to describe the brain morphology in these groups. It has been shown, moreover, that among species with convoluted brains, marine mammals (Cetacea) form a subgroup in that the cerebrocortical surface in these animals is more folded than in terrestrial mammals of similar brain size. Thus it appears that the use of a single allometric relation in studying the effect of size increase on the geometry of the brain – irrespective of convolutedness or ecological strategy – as has been done in previous studies, is unjustified and obscures the differences in the structural organization of mammalian brains. Finally, a dimensionless index of cortical folding is proposed, which indicates that the degree of cortical folding depends not only on the volume of the brain or cortex but on cortical thickness as well.

Introduction

The evolutionary expansion of brain size, and in particular selection pressures toward enlargement of the cerebral cortex, has been most prominent among mammals [*Mangold-Wirz*, 1966; *Jerison*, 1973; *Macphail*, 1982]. Together with its impressive size increase the dimensions of the cerebral cortex have changed, namely by becoming folded. *Baillarger* [1845] was the first to study this folding of the cortex from a quantitative point of view, by comparing the surface area of the cortex with

[1] This paper is dedicated to the late Professor Hans Elias (1907–1985).

the volume of the entire hemisphere in different animals, assuming geometric similarity of the brain.

Allometric studies, however, have revealed that the surface area of the cerebral cortex is almost a *linear* function of the volume of the brain, rather than scaling with the two-thirds power of geometric similarity [*Schlenska,* 1969, 1974; *Elias and Schwartz,* 1971; *Jerison,* 1979, 1982; *Passingham,* 1981; *Hofman,* 1982a, b; *Prothero and Sundsten,* 1984]. Consequently, the ratio of the cortical surface area to the volume of the brain (or cerebral hemisphere) should change very little among mature mammals as well as throughout individual development. From these findings, general conclusions have been drawn regarding the geometry of the brain. The present study is an inquiry into the assumed uniformity of the cerebral cortex by describing its morphology in mathematical terms. For this purpose the size range is extended by one order, and more species with lissencephalic brains not present in previous analyses have been included. Furthermore, the cerebral cortex is sharply defined, and gray and white matter are distinguished.

The models presented here are purely mathematical constructs and do not consider the mechanisms for cortical folding, nor do they specify the convolutional pattern of the cortex. Various mechanical models have been proposed with respect to these latter phenomena. *Richman* et al. [1975] and *Evrard* et al. [1978], for example, presented an elastic model for the development of (tertiary) convolutions, which is based on the differential growth of the inner and outer layers of the cortex and which suggests that the forces responsible for the development of cerebral convolutions are predominantly intracortical. *LeGros Clark* [1945] and *Todd* [1982] presented evidence for a link between the geometry of the brain and its folding pattern. *Todd's* geometric model shows that a cortex developing under the combined restraints of uniform growth and minimal radial distortion will tend to fold along lines of minimal curvature.

In addition to these mechanical models for explaining cortical folding, computer models have been developed based on empirical data in order to simulate various cortical parameters, including the cortical surface and gyral length [*Schlenska,* 1974; *Prothero and Sundsten,* 1984]. Thus, it was found that the total area of the gyral 'window' through which fibers pass from cortical neurons to the underlying white matter, is the crucial parameter for setting limits to brain size [*Prothero and Sundsten,* 1984]. Since cortical folding is considered the process that determines the gyral width, a mathematical model is presented from which the degree of cortical folding can be estimated. Whether the allometric relations and proposed models are appropriate for mammals in general, irrespective of size and evolutionary history, will be investigated. Part one of the inquiry deals with the allometry of the cortical surface, whereas the second part will consider the cortical volume.

Data Selection and Mathematical Procedures

Data Selection

Table I summarizes the principal data on brain parameters in 44 mammalian species. Data have

Table I. Size comparisons of various brain structures in mammals

Mammal	Brain volume cm^3	Prosencephalon volume cm^3	Cerebral cortex		
			volume cm^3 (gray matter)	volume cm^3 (gray + white matter)	total surface cm^2
Marsupialia					
Marmosa mitis	1.24	0.817	–	0.353	4.38
Didelphis virginiana	4.25	2.85	–	1.09	13.2
Wallabia rufogrisea	25.0	17.1	–	10.7	43.3
Macropus melanops	39.2	27.7	–	16.8	74.9
Basal insectivora					
Sorex minutus	0.106	0.0694	0.0293	0.0302	0.798
Sorex araneus	0.193	0.130	0.0540	0.0587	1.39
Crocidura russala	0.174	0.121	0.0534	0.0549	1.22
Crocidura flavescens	0.425	0.269	0.120	0.128	1.94
Setifer setosus	1.39	0.956	0.408	0.424	4.32
Erinaceus europaeus	3.23	2.17	0.967	1.06	9.24
Progressive insectivora					
Neomys fodiens	0.309	0.200	0.0880	0.102	2.12
Talpa europaea	0.985	0.634	–	0.333	3.88
Galemys pyrenaicus	1.28	0.838	–	0.491	4.93
Elephantulus fuscipes	1.28	0.812	–	0.403	4.66
Potamogale velox	3.86	2.18	–	1.34	8.28
Rhynchocyon stuhlmanni	5.89	3.85	–	2.01	14.3
Prosimii					
Tupaia glis	3.04	2.11	1.22	1.31	12.9
Galagoides demidoff	3.26	2.49	1.60	1.75	13.5
Perodicticus potto	13.5	10.4	–	7.67	39.8
Simii					
Callithrix spec.	11.5	9.17	6.09	7.60	40.4
Aotus trivirgatus	15	12	7.45	9.42	37.8
Saimiri sciureus	25.2	20.7	13.3	17.3	79.6
Cercopithecus aethiops	64.2	54.4	36	50.7	200
Macaca mulatta	100	84.8	55.1	76.2	250
Homo sapiens	1,389	1,186	683	1,167	2,430
Rodentia					
Rattus norvegicus	2.26	1.42	–	–	6.44
Oryctolagus cuniculus	7.10	4.41	–	–	13.8
Dolichotus spec.	30.8	20	13.1	16.1	65.1

Cerebral cortex		Index of folding	References
outer surface cm^2	mean thickness cm		
	-	1.00	1, 5
11.8	-	1.12	
35.2	-	1.23	
53.0	-	1.41	
	0.0367	1.00	3, 11, 12
	0.0388	1.00	
	0.0438	1.00	
	0.0619	1.00	
	0.0944	1.00	
	0.105	1.00	
	0.0415	1.00	3, 11, 12
	-	1.00	
	-	1.00	
	-	1.00	
	-	1.00	
-	-	> 1.00	
-	0.0946	> 1.00	3, 11 12
-	0.119	> 1.00	
-	-	> 1.00	
33	0.151	1.22	9
-	0.197	> 1.00	3, 11, 12
49.8	0.167	1.60	9
108	0.180	1.85	9
147	0.220	1.70	9
860	0.281	2.83	8
	-	1.00	10
12	-	1.15	9
45.9	0.201	1.42	9

been compiled from (1) *Elias and Schwartz* [1969]; (2) *Elias and Schwartz* [1971]; (3) *Frahm* et al. [1982]; (4) *Haug* [1970]; (5) *Pirlot* [1981]; (6) *Pirlot and Kamiya* [1982]; (7) *Ridgway and Brownson* [1979]; (8) *Schlenska* [1969]; (9) *Schlenska* [1974]; (10) *Stephan* [1954]; (11) *Stephan* [1961]; (12) *Stephan* et al. [1981]. The numbers before the authors' names refer to the corresponding numbers in table I. Data from the literature which are unreliable for methodological reasons, but which have nevertheless been used in previous studies, were not included in the present analysis. When only data on the weight were available, the volume was calculated by assuming a specific gravity of 1.036 g/cm^3 for fresh brains [*Frahm* et al., 1982]. In various cases, especially in some primates, carnivores and ungulates, the volume of the prosencephalon was calculated from the brain volume based upon fractions as given in the literature [*Mangold-Wirz*, 1966; *Kruska*, 1975].

Sometimes (e.g., in marsupials, insectivores and some primates) not all significant brain data for a species could be obtained from a single source. In these cases data were derived from various sources, in which brain size acted as reference variable.

In the present study the cerebral cortex includes the iso-, schizo- and palaeocortex, which means that the archicortex (i.e., hippocampus) and the primary olfactory allocortex (i.e., bulbus olfactorius and bulbus accessorius) are excluded [*Stephan* et al., 1981]. The main reason for this selection is that most of the studies on surface measurements of mammalian cerebral cortex are confined to these regions. In addition, by excluding the primary projection areas of olfaction, terrestrial and marine mammals can be better compared (since dolphins and whales have only a rudimentary olfactory system or none at all). The mean cortical thickness and the index of cortical folding were calculated from the volume-surface ratio (V_g/S_t) and the total surface-outer surface ratio (S_t/S_o), respectively. The following symbols are used in this paper: E = brain volume; V_p = prosencephalon volume; V_g = volume of the cerebral cortex (gray matter); V_{g+w} = volume of the cerebral cortex (gray + white matter); S_t = total surface area of the cerebral cortex; S_o = outer (visible) surface area of the cerebral cortex; d = mean cortical thickness. All variables are expressed in centimeter units, unless otherwise indicated.

Table I (continued)

Mammal	Brain volume cm^3	Prosencephalon volume cm^3	Cerebral cortex		
			volume cm^3 (gray matter)	volume cm^3 (gray + white matter)	total surface cm^2
Carnivora					
Felis catus	31.8	23.8	15.3	20.5	83.3
Procyon lotor	42.6	–	–	–	122
Vulpes vulpes	48.3	37.1	27.4	34.4	151
Canis latrans	85.3	68.2	–	–	227
Ungulata					
Ovis aries	105	83.5	51.8	71.6	298
Sus scrofa	162	126	78.8	110	360
Bos taurus	486	389	226	337	1,055
Equus caballus	609	487	282	454	1,228
Proboscidea					
Loxodonta africana	4,148	2,815	1,402	2,491	6,275
Cetacea					
Phocaena phocaena	483	378	205	340	1,300
Stenella styx	623	510	–	455	1,474
Delphinus delphis	722	–	–	–	1,717
Tursiops truncatus	1,118	884	463	815	2,700
Grampus griseus	1,588	1,205	624	1,127	3,132
Globicephala macrorhyncha	2,786	2,184	1,177	2,045	5,815
Pseudorca crassidiens	3,650	–	–	–	7,392

For the sources of data see section 'Data Selection'; dashes indicate that no data were available. The cerebral cortex includes neo-, schizo-, and palaeo-cortex. The index of cortical folding is defined as S_t/S_o.

Mathematical Procedures

In order to describe the statistical relationship between two random variables the allometric relationship $\log y = \log a + \alpha \log x$ is generally used, in which coefficient α is the slope while log a is the y-intercept at $x = 1$. Since in most cases the variables are subject both to biological fluctuations and to errors of measurement, the standard (or reduced) major axis (also called the geometric mean regression line) is preferred to the conventionally used method of linear regression [*Ricker*, 1984]. An objection raised against the use of this method [*Jolicœur and Heusner*, 1971] in favor of the ordinary major (= principal) axis, however, is that the slope of the standard major axis is almost insensitive to the strength of the relationship, as reflected by the Spearman correlation coefficient r [*Hofman* et al., 1986]. However, as the correlation (r) approaches unity, as in the present study ($r > 0.99$), the standard major axis and the principal axis will converge. The standard major axis, in addition, is symmetrical and invariant with changes of scale. Therefore, and for reasons of com-

Cerebral cortex		Index of folding	References
outer surface cm^2	mean thickness cm		
52.3	0.184	1.59	9
66	-	1.85	1
83.7	0.181	1.80	9
126	-	1.80	1, 2
154	0.174	1.94	9
166	0.219	2.17	
424	0.214	2.49	
439	0.230	2.80	
1,634	0.223	3.84	4
315	0.158	4.13	4
-	-	> 1.00	6, 7
430	-	3.99	1
567	0.171	4.76	1, 4
730	0.199	4.29	4
1,110	0.202	5.24	1, 4
1,488	-	4.97	1

putational simplicity, the method of the standard major axis has been used in this study [for statistic see *Jolicœur and Mosimann*, 1968; *Clarke*, 1980; *Ricker*, 1984; *Hofman* et al., 1986].

Size and Shape of the Cerebral Cortex

The gross morphology of the brain and cerebral cortex in mammals, ranging from the insectivore *Sorex minutus* ($E = 0.106\ cm^3$; $S_t = 0.8\ cm^2$) to the elephant *Loxodonta africana* ($E = 4{,}148\ cm^3$; $S_t = 6{,}275\ cm^2$), has been described in mathematical terms. Table II summarizes the results of the allometric analysis [i.e., the scaling factors (α) and their associated statistics] of the outer surface area (S_o) and the total surface area (S_t) for various brain parameters. It appears that mammals can be subdivided into two major groups: (1) small mammals with lissencephalic brains, where surface-volume relationships are determined by linear geometric laws [surface $\propto$ (volume)$^{2/3}$], and (2) mammals with gyrencephalic brains, where strict geometric similarity no longer holds. The size of the cortex at which the transition from a lissencephalic to a gyrencephalic brain takes place, i.e., at which convolutions appear at the cortical surface, is at a surface area of about $10\ cm^2$, in other words when $E = 3\text{-}4\ cm^3$ and $V_g = 1\text{-}1.5\ cm^3$ (fig. 1).

For species in which the cortex is convoluted, i.e., where $S_t > S_o$, a further distinction can be made between terrestrial and marine mammals, i.e., dolphins and whales (walruses, seals and other semi-marine mammals were not included in the present study, nor were aerial species). For terrestrial mammals the relationship between the total and outer cortical surface ($S_t > S_o$) is given by:

$$S_t = 0.566\ S_o^{1.25} \qquad (1)$$

($r = 0.999$; 95% confidence limits for $\alpha = 1.21\text{-}1.28$). A similar relationship can be derived for dolphins and whales (table II). Although the slopes of the regression

Table II. Allometric analysis of the cerebral cortex in mammals

Variate	Folding index = 1		Folding index > 1			
			terrestrial mammals		marine mammals	
	S_o	S_t	S_o	S_t	S_o	S_t
Outer cortical surface, cm^2	-	1.00	-	1.25 ± 0.015 (19)	-	1.16 ± 0.054 (6)
Total cortical surface, cm^2	1.00	-	0.801 ± 0.009 (19)	-	0.863 ± 0.039 (6)	-
Cortical volume (gray), cm^3	0.668 ± 0.041 (7)		0.731 ± 0.015 (13)	0.890 ± 0.017 (16)	0.726 ± 0.018 (4)	0.848 ± 0.046 (4)
Cortical volume (total), cm^3	0.638 ± 0.023 (12)		0.659 ± 0.016 (16)	0.820 ± 0.018 (21)	0.704 ± 0.012 (4)	0.849 ± 0.053 (5)
Prosencephalon volume, cm^3	0.672 ± 0.023 (13)		0.715 ± 0.015 (17)	0.887 ± 0.018 (22)	0.721 ± 0.013 (4)	0.874 ± 0.055 (5)
Brain volume, cm^3	0.659 ± 0.024 (13)		0.728 ± 0.022 (18)	0.901 ± 0.022 (23)	0.748 ± 0.024 (6)	0.877 ± 0.034 (7)

The slopes (standard major axis ± SD) of the regression lines have been derived from the allometric equation $\log y = \log a + \alpha \log x$, where x is the variate, y the covariate (here S_o and S_t) and log a is the y-intercept. Figures in parentheses refer to the number of species used. S_o = Outer surface area of the cerebral cortex; S_t = total surface of the cerebral cortex. The allometric equations are all significant at the 0.1% level, with $r > 0.99$.

lines vary, the differences are not statistically significant (d.f. = 12.5, t = 1.79; $p > 0.05$), implying that the total cortical surface scales to the outer cortical surface in a similar way in both groups. It therefore appears that the distinction between terrestrial and marine mammals is not a scaling disparity, as in convoluted versus non convoluted brains, but rather reflects a difference between the regression constants, as a result of which the regression line in marine mammals is shifted upwards (d.f. = 23, t = 3.07, $p < 0.01$; see fig. 1). Thus, the total surface area of the cortex in marine mammals is about 60% larger than it would be in a terrestrial mammal of a given outer surface area.

When we consider the cerebral cortex of terrestrial mammals in more detail we find that the cortical surface area varies with brain volume as follows:

$$S_o = 4.40\ E^{0.728} \qquad (2)$$

($r = 0.993$; 95% confidence limits for $\alpha = 0.685$–0.776) and

$$S_t = 3.77\ E^{0.901} \quad (3)$$

($r = 0.994$; 95% confidence limits for $\alpha = 0.857$–0.947). Note that whereas in nonconvoluted brains $S_t \propto E^{0.66}$, in brains where the cortex is folded $S_t \propto E^{0.90}$. It thus appears that the use of a single allometric relation in studying the effect of size increase upon the geometry of the brain, irrespective of convolutedness or ecological strategy [as has been done in previous studies, e.g., *Elias and Schwartz,* 1971; *Hofman,* 1982a; *Jerison,* 1983; *Prothero and Sundsten,* 1984] is unjustified and obscures the differences in the structural organization of mammalian brains. Discrepancies between observed and expected cortical surface areas will be most prominent in extremely small animals with lissencephalic brains, since the *general* mammalian regression line for the relationship between the total surface area and brain volume ($\alpha = 0.897 \pm 0.012$, cf. equation 3) hardly deviates from that of terrestrial mammals with gyrencephalic brains ($t = 0.071$; $p > 0.45$).

Analysis of the intrinsic geometry of the cerebral cortex shows that the surface-volume relationships in (terrestrial) mammals with lissencephalic brains ($S_t = S_o$) is described by:

$$S_t = a_1\ V_g^{0.668} \quad (4)$$

($r = 0.991$; 95% confidence limits for $\alpha = 0.572$–0.780) and in terrestrial mammals with gyrencephalic brains by:

$$S_t = a_2\ V_g^{0.890} \quad (5)$$

($r = 0.997$; 95% confidence limits for $\alpha = 0.851$–0.931). Note that, since $S_t = d^{-1}\ V_g$ by definition, where d is the mean cortical thickness, it follows that $a_1 = d^{-1}\ V_g^{0.33}$ and $a_2 = d^{-1}\ V_g^{0.11}$. From these equations it becomes evident that the cortical thickness in lissencephalic brains is proportional to the cube root of the cortical volume, whereas in animals with gyrencephalic brains the mean cortical thickness increases according to $d \propto V_g^{0.11}$. Denoting cortical thickness in nonconvoluted brains by d_1 and in convoluted brains by d_2, it appears that $d_2 \propto \sqrt[3]{d_1}$ in terrestrial mammals. In conclusion we may say that, whereas in small-brained species the cortical volume expands by virtue of a combined increase in surface area and cortical thickness, the increase of the cortical volume in species with a convoluted cortex is almost entirely

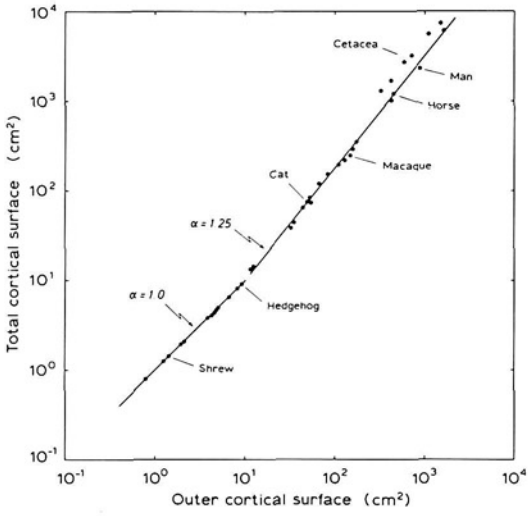


Fig. 1. Total cortical surface area as a function of outer cortical surface area. Logarithmic scale. Three groups can be distinguished: (1) Mammals with lissencephalic brains ($\alpha = 1$); (2) terrestrial mammals with gyrencephalic brains ($\alpha = 1.25$; see equation 1), and (3) marine mammals (dolphins and whales) with gyrencephalic brains. Statistics of the regression lines are given in table II.

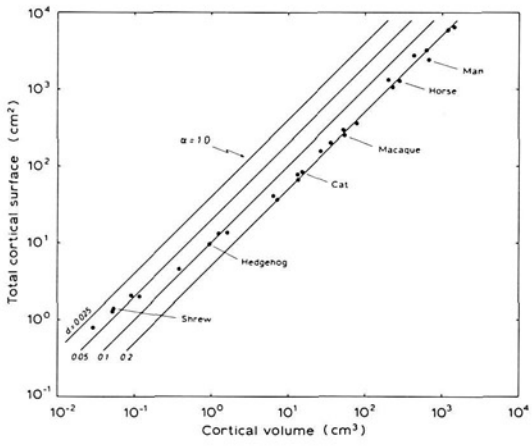


Fig. 2. Total cortical surface area as a function of cortical volume. Logarithmic scale. The lines denote values of cortical thickness (0.025, 0.05, 0.1 and 0.2 cm). The figure shows that whereas the mean cortical thickness at first increases as a function of cortical volume, it seems to reach an asymptotic value at about 0.25 cm. Notice that the human cortex is relatively thick.

due to a disproportionate expansion of the cortical surface area (fig. 2).

Scaling Model of Cortical Folding

The index of folding (FI) of the cerebral cortex in mammals is expressed as the ratio of total surface area to outer surface area. Since the total cortical surface area is expressed as $S_t = d^{-1} V_g$, only the outer surface-volume relationship needs to be determined in order to calculate the index. Allometric analysis shows that $S_o \propto V_g^{0.67}$ for nonconvoluted brains and $S_o \propto V_g^{0.73}$ for convoluted brains (see table II). To arrive at a dimensionless index of cortical folding, in which cortical volume is the principal variable, the S_o equations have to be dimensionally balanced, like the S_t equation. Although this condition is met for nonconvoluted brains, this is not the case with convoluted brains. In previous studies, however, it has been found that the thickness of the cortex is an important factor, as well, in determining the degree of cortical folding [*Hofman*, 1983; *Prothero and Sundsten*, 1984]. It seemed plausible, therefore, to introduce cortical thickness into the scaling model as an additional parameter. According to this model, the surface-volume relationships in terrestrial mammals can be described by:

$$S_o = a_3 V_g^{2/3} \quad (S_t = S_o) \tag{6a}$$

and

$$S_o = a_4 (d^{-1} V_g^3)^{1/4} \quad (S_t > S_o) \tag{6b}$$

Comparison between the multivariate model (equation 6b) and the bivariate model, furthermore, revealed that the residual variance diminishes by introducing d into the analysis. In marine mammals, however, introduction of cortical thickness into the analysis did not increase the strength of the relationship; hence, the scaling model is confined to the bivariate relation:

$$S_o = a_5 V_g^{3/4} \quad (S_t > S_o) \tag{6c}$$

When $S_t = d^{-1} V_g$ is subsequently divided by S_o (equation 6), with regression constant a being empirically determined, the index of cortical folding can be derived. For lissencephalic brains, where

$$FI = 0.115\, d^{-1} V_g^{1/3} \tag{7a}$$

the folding index is a constant, and equal to one by definition. In terrestrial mammals with gyrencephalic brains ($S_t > S_o$) the folding index is given by:

$$FI = 0.20\,(d^{-3}\,V_g)^{1/4} + 0.1 \qquad (7b)$$

and in marine mammals ($S_t > S_o$) by:

$$FI = 0.22\,d^{-1}\,V_g^{1/4} - 1.2 \qquad (7c)$$

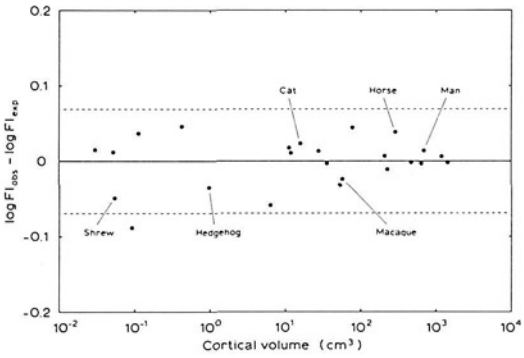


Fig. 3. Deviation between the observed and expected index of cortical folding (FI) as a function of cortical volume. The expected indices were calculated from equation 7. With the exception of the water shrew *(Neomys fodiens)* the deviations between observed and estimated indices are less than ± 15%, indicated by the dashed lines. Notice that the reliability of the estimate is independent of the size of the cortex.

According to these models, which are revised and extended versions of a nonlinear geometric model developed in an earlier study [*Hofman,* 1983], cortical folding depends not only upon the volume of the brain or cortex but on cortical thickness as well. Although the increase in cortical thickness among mammals is not more than one order of magnitude, its significance as a brain parameter lies in the fact that it reflects the structural complexity of the cortex [*Jerison,* 1973; *Powell,* 1981; *Hofman,* 1985]. It has been shown, for example, that the reciprocal of cortical thickness is a linear function of the neuron density [*Hofman,* 1982b]. This means that a decrease in neuron density, caused by an increase in cell size and number of interconnecting nerve fibers, leads to a reduction in the degree of cortical folding. A reduction of cell layers, on the other hand, as in dolphins and whales [*Morgane* et al., 1983] may lead to a relatively more convoluted cortex.

That the allometric equations can be used for predictive purposes as well follows from the variance analysis of the deviations between observed and estimated indices of cortical folding, which were found to be less than ± 15% (fig. 3). Another indication of the goodness of fit between the model and the empirical findings is that, in addition to the high explained percentage of variation, the residuals are randomly distributed. These models, therefore, can be used as reliable estimators in cases where parametric values, such as total cortical surface area, cortical thickness and degree of cortical folding are not available, as in fossil brains. The models, however, generally require specific knowledge of the size of the cortical volume before other cortical parameters can be estimated. Since it is not always possible to determine the volume of the cerebral cortex by direct measurement, it would be useful to estimate its volume indirectly, for example, through a gross variable such as total brain size. These relationships will be considered in the second part of the inquiry.

Concluding Remarks

Among the most distinctive morphological features of vertebrate brains is the evolutionary expansion of the cerebral cortex. The cortex is characterized by a compact laminar organization of cells and fibers, with local morphological differences. These differences - which are due partly to phyletic differences, and partly to the degree to which the species in question is adapted to make use of various cortical subdivisions in its interaction with its environment - are of degree rather than of kind, and seem to exist within an overall organizational framework that is common to all vertebrates [*Hodos,* 1982; *MacLean,* 1982]. In mammals, furthermore, the similarities in mode of organization of the cerebral cortex have been found to be more striking than the differences, among cortical subdivisions as well as among species [*Mountcastle,* 1978; *Powell,* 1981]. The present analysis provides a plausible theoretical basis for the assumed uniformity of the cortex, by showing that both the growth and convolutional development of mammalian cortex can be described by simple geometric models. The model for convoluted brains predicts that large (and therefore usually more advanced evolutionarily) brains tend to be more highly convoluted than small brains, which is due to the disproportionately small changes in cortical thickness. This finding is in agreement with a mechanical model of cortical folding, which predicts that the intergyral wavelength is a function of the cortical thickness [*Richman* et al., 1975; *Todd,* 1982; *Prothero and Sundsten,* 1984]. Moreover, it explains the high degree of cortical folding found in marine mammals. The fact is that in dolphins and whales the most advanced anatomical features of the cerebral cortex are regarded as characterizing a conservative protoneocortical stage in evolution [*Kesarev* et al., 1977; *Morgane* et al., 1983; *Bullock,* 1984]. It has been found, for instance, that the cortex between proisocortices shows no indications of hypergranular cores. Consequently, the brain in marine mammals retains all conservative characters seen in primitive terrestrial and aerial forms, such as hedgehogs and bats [*Schwerdtfeger* et al., 1984; *Morgane* et al., 1985]. It seems that the final stage of neocortical evolution, as seen in most terrestrial mammals, has apparently not been reached in dolphin and whale brains. An index of corticalization, therefore, as has been proposed for mammals in general [*Hofman,* 1985], based upon a measure of relative cortical surface area, has to be used with some caution when applied to marine mammals.

Although the evolution of the cerebral cortex was accompanied by the development of more differentiated cell types and coupling functions, and by the reorganization of neural structures [*Ebbesson,* 1984; for a review see *Northcutt,* 1984] corticalization in mammals is mainly characterized by the development and multiplication of more specialized neuronal circuits [*Mountcastle,* 1978; *Hofman,* 1985]. Since these functional units are organized in vertical columns, an increase in the number of the modules, as an adaptive response to complex environmental demands, will be reflected by an expansion of the cortical surface area. In this respect the evolution of the cerebral cortex resembles that of organisms in general, in that it seems 'largely the story of the struggle to increase surface

in proportion to volume' [*Haldane,* 1928]. According to this view, sensory information from the external world can be represented by a two-dimensional matrix. The capacity of the brain to construct 'real' worlds, in which the information from various modalities is integrated as information about objects in space and time, depends upon how the animal's brain accomplishes the processing and integration of the information. It is in this sense that 'reality' can be thought of as a creation of the brain.

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